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| **Fiscal Year:** | FY 2011 | **Task Last Updated:** | FY 05/23/2011 |
| **PI Name:** | Thomopoulos, Stavros Ph.D. | | |
| **Project Title:** | The effect of weightlessness on the tendon-to-bone insertion | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Developmental biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Animal Biology: Vertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Musculoskeletal Biology | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
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| **City:** | St. Louis | **State:** | MO |
| **Zip Code:** | 63110-1010 | **Congressional District:** | 1 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2007 Space Flight: Bion-M1 NNH07ZTT001N |
| **Start Date:** | 08/01/2009 | **End Date:** | 07/31/2013 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Austin, Edward | **Contact Phone:** | 650-604-2108 |
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| **Flight Program:** | Shuttle | | |
| **Flight Assignment:** | STS-131 | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX09AP05G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Soft tissue injuries often occur at the sites of attachment of tendons and ligament to bone, the so-called "insertion sites". The sensitivity of the musculoskeletal system to its mechanical environment may magnify the injury susceptibility of insertion sites during prolonged exposure to microgravity. Changes in loading across joints, as would occur during a long stay on the International Space Station or during a trip to Mars, will lead to changes in tissue mechanical, structural, and biochemical properties. Joint unloading will result in a catabolic biologic state and subsequent tissue degeneration. Physiologic joint loading, on the other hand, will result in an anabolic biologic state and maintenance of tissue properties. Joint deterioration during space flight would significantly increase the risk of injury to the tendons and ligaments which provide function and stability to joints. These injuries do not heal well (e.g., the failure rate for repaired rotator cuff repairs may be as high as 94%) and will severely debilitate an injured astronaut. The negative effect of unloading has most clearly been demonstrated in bone and muscle. Less is known, however, about the effect of weightlessness on joint soft tissues. Therefore, the overall objective of this proposal is to examine the effect of prolonged weightlessness on the mechanical and biochemical properties of muscles, tendons, and their insertions into bone. We propose to study the tendon-to-bone insertions of animals subjected to fourteen to thirty days of weightlessness through the Biospecimen Sharing Program organized for the STS-131, STS-133, and the Bion-M1 missions. These shared animals will allow us to examine joint degeneration due to weightlessness by comparing the weightless group to a control group on Earth. Outcome measures will include assessment of the mechanical, structural, genetic, and compositional changes of mouse tendon-to-bone insertions. These analyses will be performed on mice whose procedures and objectives were associated with the primary research focus of this research program. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Musculoskeletal injuries are a common cause of pain and disability, and result in significant health care costs. Approximately 30% of work related injuries in the United States each year are to the upper extremity. At the rotator cuff, tendon-to-bone repair techniques have been plagued by a high rate of recurrent tears, ranging from 20-94% There is little scientific basis for the current clinical management of this injury. Regeneration of the insertion site of the tendon is critical for the restoration of tendon function and for the prevention of re-injury. However, clinical and animal model studies have demonstrated a lack of healing potential after tendon-to-bone injury and repair.<br>Musculoskeletal tissues are sensitive to their mechanical environment. Increased loading (e.g., through exercise) generally leads to an increase in mechanical properties, while decreased in loading (e.g., through immobilization) generally leads to decreases in mechanical properties. For example, bone adapts to its loading environment by remodeling its architecture to optimally transfer stresses. Unloading leads to rapid resorption of bone and a decrease in bone mineral density. Increased loading, on the other hand, promotes new bone formation and an increase in bone mineral density. These changes are especially evident at the bony end of the tendon-to-bone insertion. Understanding the responses of tendons and their bony insertions to a weightless environment will allow us to develop strategies for preventing tendon-to-bone injuries and/or improving the healing response of tendon-to-bone injuries for humans on Earth. |
| **Task Progress:** | 1.The effect of microgravity on gene expression in the humerus<br>Joint unloading on earth or in space has been shown to cause a rapid loss of bone mineral mass. Changes in mineral are typically due to increased bone resorption by osteoclasts, resulting in decreased mineral content as well as changes in the trabecular bone architecture. It is likely that similar bony changes will occur at tendon-to-bone insertions subjected to a weightless environment. Preliminary data from the STS131 BSP demonstrate that bone resorption is increased (as indicated by increases in the ratio of RANKL and OPG gene expression) and osteoblast differentiation is decreased (as indicated by decreases in OSX gene expression). Surprisingly, adipogenesis was decreased due to the weightless environment (as indicated by decreases in PPARG gene expression). As expected, the mechanoresponsive gene COX2 was dramatically downregulated due to microgravity.<br><br>2. The effect of microgravity on the rotator cuff muscles<br><br>Microgravity significantly downregulated the expression of both myogenic and adipogenic. MyoD is involved in initiating a myogenic differentiation program of multipotential stem cells and myostatin is a negative regulator of muscle development. The mRNA changes indicate atrophy/degeneration of the muscles due to microgravity. C/EBPa mediates transcriptional control of adipogenesis. Upregulation in its expression has been observed due to muscle unloading. Leptin is manufactured in fat cells and its upregulation suggests presence of adipose tissue. Surprisingly, we found downregulation of adipogenic markers, indicating that degeneration due to microgravity may occur through different pathways than degeneration due to tendon and/or nerve injury. Microgravity also led to an increase in collagen type III, a marker of fibrosis (and hence degeneration) in muscle. Interestingly, effects were muscle specific; the infraspinatus was most sensitive to the unloading conditions in general, but only the supraspinatus showed changes in adipogenesis.<br><br>3. The effect of microgravity on the rotator cuff tendons and their bony insertions<br><br>Joint unloading through cast immobilization or muscle paralysis results in a catabolic biologic state and subsequent tissue degeneration. For tendons and their bony insertions, enzymatic degradation of the tissue by the matrix metalloproteinases (MMPs) is most evident as changes in the extracellular matrix; collagen fibers lose their characteristic organization and proteoglycans will be digested. Joint unloading due to a weightless environment led to similar changes, with significant increases in MMP-3 and MMP-13 mRNA expression and significant decreases in tendon cross-sectional area. However, for the time course of spaceflight studied here, the gene expression and geometric changes to tendons and their bony insertions did not result in statistically significant changes in tissue mechanical properties. |
| **Bibliography Type:** | Description: (Last Updated: 10/19/2020) |
| **Abstracts for Journals and Proceedings** | Lim C,Shen H, Kotiya A, Silva MJ, Thomopoulos S. "STS-131: The Effects of Microgravity on the Rotator Cuff of the Shoulder." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011.<br>18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011 |

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| **Abstracts for Journals and Proceedings** | Lim C, Shen H, Thomopoulos S. "Microgravity Leads to Degeneration of Murine Rotator Cuff Muscles." 57th Annual Meeting of the Orthopaedic Research Society, Long Beach, CA, January 13-16, 2011.<br>Transactions of the 57th Annual Meeting of the Orthopaedic Research Society, 2011 Feb;57:208. , Feb-2011 |